

Cyclic Loading of PBX 9501 and PBX 9502: Compression and Tension

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Abstract. The uniaxial mechanical properties of plastic-bonded explosives (PBX) 9501 and 9502 have been studied extensively over the past 25+ years. Uniaxial, cyclic loading compression and tension tests were performed on PBX 9501 and PBX 9502 spanning a range of temperatures and loading rates. All cyclic tests were analyzed to quantify damage accrual within a specimen as a function of increasing strain for each cycle. One full cycle is defined as the loading, unloading, and relaxation time at zero load. The damage accumulation for each test condition is evaluated using two parameters. The first is the irreversible strain, or residual strain, measured after each full cycle, i.e. at the end of the relaxation time before loading of the next cycle begins. The second is the change in modulus, or stiffness, of each loading portion for each cycle. In the work presented, damage accrual of PBX 9501 and PBX 9502 are determined empirically for both compression and tension cyclic loading experiments.

INTRODUCTION

The plastic-bonded explosive (PBX) 9501 and 9502 composite materials are composed of granular explosive crystals surrounded by a polymer binder. PBX 9501 is an HMX-based material and PBX 9502 is a TATB-based material [1]. These two materials have been thermal-mechanically characterized extensively for over 25 years, including via uniaxial compression and tension experiments, typically monotonic tests to failure [2-5]. In recent years uniaxial cyclic loading experiments in compression and tension have been performed on PBX 9501 and PBX 9502 to provide insight into complex loading histories that exhibit the distinction between reversible (elastic) and irreversible strain in the form of plasticity and damage within the material [6].

SPECIMEN MANUFACTURE

All specimens were machined from isostatically pressed hemispherical charges. Tension specimens have an overall length of 3.0 inches, a gauge length of 1.5 inches long and 0.5 inches in diameter in the center of the specimen where measurements are made, and a 15° taper at both ends to fit into the tensile grips. Compression specimens are right cylinders that are 1.125 inches long by 0.375 inches in diameter. Density of all specimens is measured using immersion methods. PBX 9501 exhibited densities of 1.8362 ± 0.0005 g/cm³ for tension specimens and 1.8395 ± 0.0080 g/cm³ for compression specimens. For PBX 9502, immersion density measurements reported to be 1.8933 ± 0.0005 g/cm³ and 1.8899 ± 0.0012 g/cm³ for tension and compression, respectively.

EXPERIMENTAL PROCEDURE

All tests were performed on an Instron 5567 retrofitted with an MTS controller. Strain on the specimen was measured with two oppositely mounted knife-edge extensometers, Instron model # 826-2026. Extensometers have

curved blades slightly larger than 0.5 inches to minimize point loading on the specimen. Test temperature is applied by a Bemco, Inc. environmental chamber and controlled to within $\pm 0.2^\circ\text{C}$ of the desired test temperature.

All cyclic tests have a total of nine cycles. A cycle is defined to have three phases: a loading phase, an unloading phase, and a relaxation phase. Each loading phase incrementally increases the crosshead displacement for each cycle. All relaxation phases are 2 hours long for compression and 1 hour long for tension tests. For tension tests, a 1 hour hold at the top of each cycle is included. The strain for tension tests of these explosive composites is very low for monotonic tests. To exaggerate the strain evolution for each cycle a creep hold is included for irreversible strain to evolve with each cycle. This is not needed for compression tests. Tests are performed at multiple temperatures and multiple test speeds for each stress state (compression and tension) for each material with one to two tests per condition.

Raw stress versus strain curves for all PBX 9501 compression and tension tests are shown in in the top of Figure 1, left and right, respectively. All PBX 9502 stress versus strain curves for compression and tension are plotted at the top in Figure 2, left and right, respectively. All tests are plotted again as strain versus time below corresponding stress-strain data in each figure. In the stress-strain plots in the top of Figures 1 and 2 the temperature and rate dependence of both materials is seen. As temperature decreases and/or rate increases the material is stronger and stiffer than tests performed at warmer and/or slower loading rates. This is typical of PBXs and has been seen previously in monotonic loading experiments [2-5].

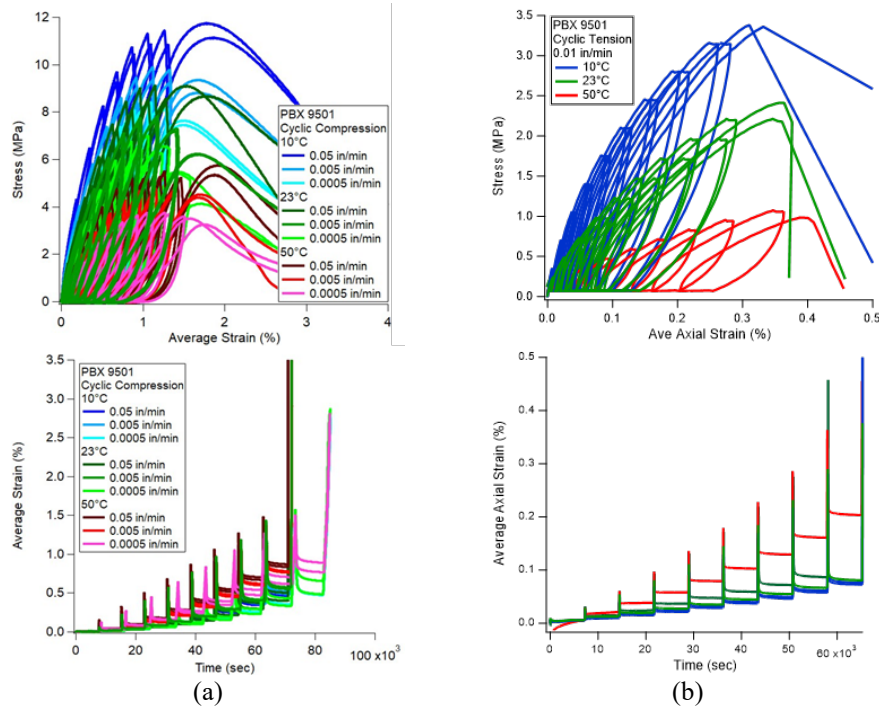


FIGURE 1. Stress vs. strain (top) and strain vs. time (bottom) for compression (a) and tension (b) of all PBX 9501 cyclic loading tests at multiple temperatures and test speeds. See legend for conditions.

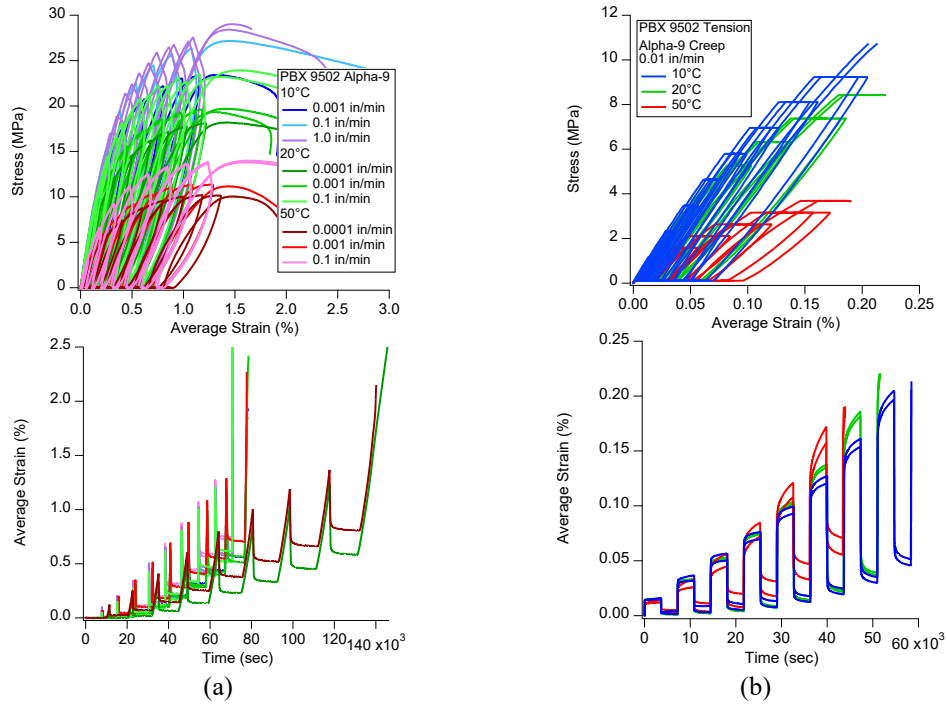


FIGURE 2. Stress vs. strain (top) and strain vs. time (bottom) for compression (a) and tension (b) of all PBX 9502 cyclic loading tests at multiple temperatures and test speeds. See legend for conditions.

Data Analysis

For both tension and compression, we investigate the effect of load history on the material. In the analysis that follows we attempt to quantify effects of plastic deformation and damage in compression and tension. For this we use a simplified empirical analysis of the cyclic loading data that allows us to quantify irreversible strain and damage in these specimens. First, the test data are divided into individual cycles and then the initial time and strain values of each cycle are set to zero. An example of this is shown in Figure 3 for compression and tension on the left and right, respectively, with stress versus strain on the top and strain versus time on the bottom.

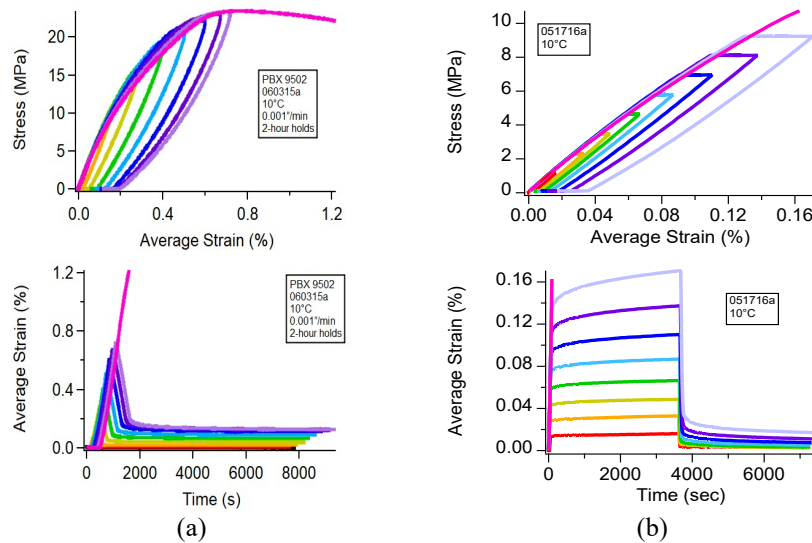


FIGURE 3. Example of individual cycles (color-coded for cycle number) for compression (a) and tension (b) after initial time and strain values have been set to zero. Stress vs. strain (top) and strain vs. time (bottom).

Irreversible strain is defined to be the remaining strain after one cycle. In the average strain versus time plots of Figure 3 this is effectively the strain value at the far right of the curve for each cycle (note that accumulated irreversible strain from the previous cycles have already been subtracted from the Figure 3 curves, so irreversible strain is for that cycle only). This measure of irreversible strain is an approximation. Longer relaxations would lead to lower values. Somewhat arbitrarily, the modulus (stiffness) measurement is defined to be the slope of the stress versus strain curve at approximately 10% of the total stress maximum for both compression and tension.

RESULTS AND DISCUSSION

PBX 9501: Compression and Tension

Figure 4 shows the compressive and tensile irreversible strain and modulus at 10% of ultimate stress versus the maximum strain applied during each cycle for all temperatures and loading rates for all PBX 9501 tests. Each point on the graph in Figure 4 is the average for conditions with more than one specimen. The irreversible strain plot in Figure 4 shows continual increase with each cycle. This is expected as the irreversible strain may contain contributions from the viscous (time-dependent) nature of the elastic deformation, as well as the plastic (non-recoverable) deformation arising from damage in the material as each cycle displacement increases. The irreversible strain is higher for the hot tests, as expected because the binder is softer at elevated temperatures. As temperature decreases the irreversible strain decreases as the binder becomes firmer. There is one test condition from the compression experiments that falls in the middle and appears to be a transition in the data between the two irreversible strain camps. It is unknown why this transition exhibits itself at this temperature and loading rate. For tension we also see a difference between the hot data and the other temperatures. Due to the limited number of tensile specimens, only one loading rate was performed at each temperature. Therefore, we don't see the same transition for compression with the limited tensile data.

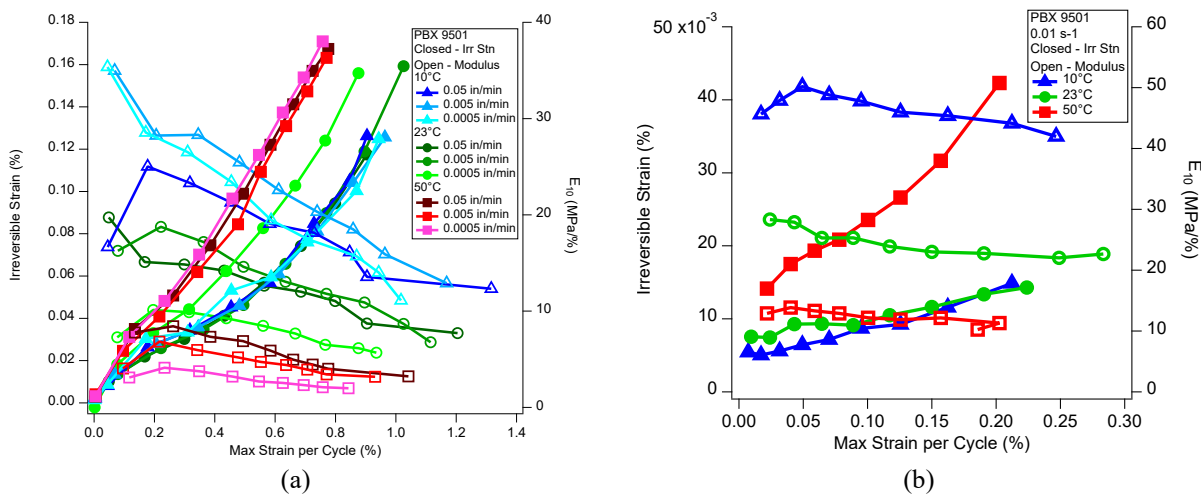


FIGURE 4. PBX 9501 compressive (a) and tensile (b) parameters of irreversible strain (%) (left axis, solid symbols) and modulus at 10% of ultimate stress (MPa/%) (right axis, open symbols) plotted vs. the maximum strain value of each cycle. The responses at 50°C (red shades), 23°C (green shades), and 10°C (blue shades) and multiple rates (slower to faster, lighter to darker) are compared.

The modulus data in Figure 4 shows an increase from the first cycle to the second cycle for most test conditions. This may be an indication of the compression specimen settling against the upper and lower platen or the tensile specimens settling into the grips in the low-load region where little to no damage has begun to occur. All tests decrease in stiffness with each subsequent cycle, corresponding to the damage of the material. Tests performed at lower temperatures exhibit more overall modulus degradation than tests performed at higher temperatures. Although the material exhibits more irreversible strain (plasticity) at higher temperatures it appears that damage accrual, evaluated via modulus degradation, is higher as temperature decreases.

PBX 9502: Compression and Tension

Figure 5 shows the compressive and tensile irreversible strain and modulus at 10% of ultimate stress versus the maximum strain applied during each cycle for all temperatures and loading rates for all PBX 9502 tests. Each point on the graph in Figure 5 is the average for conditions with more than one specimen. The irreversible strain plot in Figure 5 shows continual increase with each cycle. This is expected as the irreversible strain may contain contributions from the viscous (time-dependent) nature of the elastic deformation, as well as the plastic (non-recoverable) deformation arising from damage in the material as each cycle displacement increases. For compression the irreversible strain curves are very similar for all temperatures and rates. For tension, the irreversible strain is higher for the hot tests, as expected because the binder is softer. As temperature decreases the irreversible strain decreases as the binder becomes firmer. It is unknown why this transition exhibits itself at this temperature and loading rate. One possibility may be related to the glass transition temperature (T_g) of the Kel-F binder, which is at $\sim 28^\circ\text{C}$ [1]. In tension, the binder plays more of a role in the mechanical response whereas the amount of solids loading of the composite plays more of a role in compression. Therefore, it is possible that tests performed on either side of the T_g might exhibit different behavior. However, this is just a hypothesis.

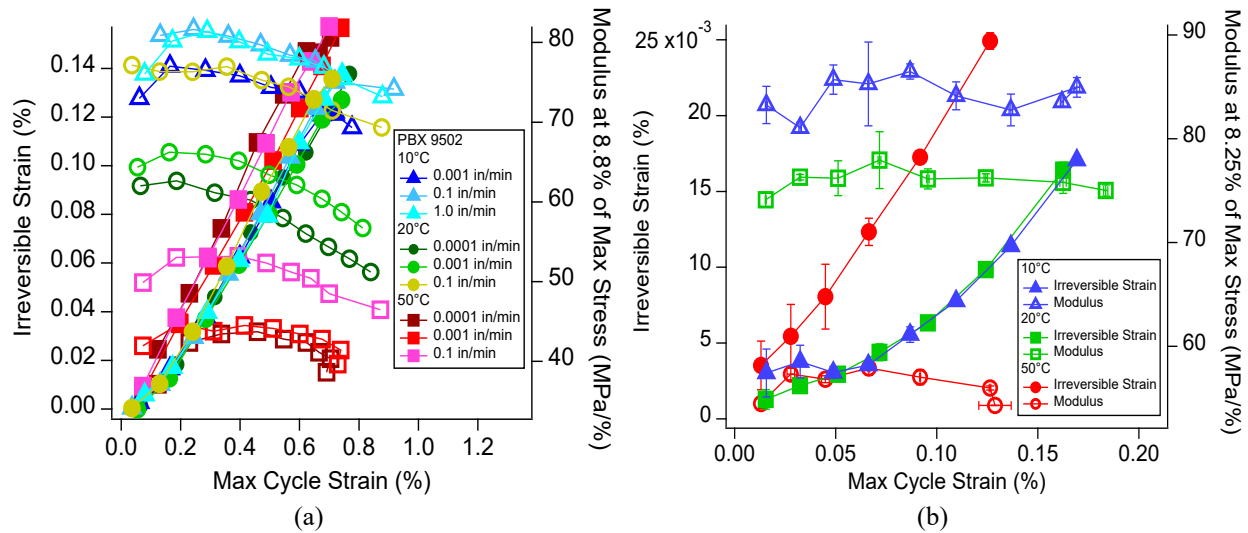


FIGURE 5. PBX 9502 compressive (a) and tensile (b) parameters of irreversible strain (%) (left axis, solid symbols) and modulus at 10% of ultimate stress (MPa/%) (right axis, open symbols) are plotted vs. the maximum strain value of each cycle. The responses at 50°C (red shades), 20°C (green shades), and 10°C (blue shades) and multiple rates (slower to faster, lighter to darker) are compared.

The modulus values in Figure 5 show an increase from the first cycle to the second cycle for most test conditions. This may be an indication of the compression specimen settling against the upper and lower platen or the tensile specimens settling into the grips in the low-load region where little to no damage has begun to occur. For compression, all tests decrease in stiffness with each subsequent cycle, corresponding to the damage of the material. For tension, this change is slight with the modulus values changing less than 5% from the initial modulus. This is likely because the magnitude of the tensile parameters is much smaller than in compression; in tension failure is flaw-driven and the material cannot be taken out very far in strain before an initiation point of damage has coalesced into a full fracture.

CONCLUSION

PBX 9501 and PBX 9502 were cyclically loaded in compression and tension at various temperatures and rates. The data were analyzed empirically by investigating and comparing the irreversible strain and modulus changes for each cycle. For PBX 9501 the irreversible strain evolution in compression is similar for most tests performed at ambient and cold with hot tests exhibiting more irreversible strain with a different curve shape. For PBX 9501 tension

tests we see similar temperature dependence for irreversible strain where hot tests have higher and more irreversible strain than the ambient and cold tests. For PBX 9502, the irreversible strain is similar for all compression tests with little difference between temperatures. However, for tension, the irreversible strain evolution is similar for tests performed at ambient and cold but hot tests exhibit more irreversible strain. The difference is attributed in hot tension tests being above the Kel-F T_g of 28°C where the binder flows more. This is not seen in the PBX 9502 compression data because compressive parameters rely more on the solids loading of the composite formulation, whereas tension parameters rely more on the binder properties. Generally, for both PBX 9501 and PBX 9502, modulus values increase slightly for the first cycle (or a few for tension) then begin to decrease, which is attributed to the damage accumulation in subsequent cycles. Modulus values also exhibit temperature and rate dependence, as expected.

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